Toy World – E-commerce-Database-Management-System

SQL Query Implementation for E-Commerce Database

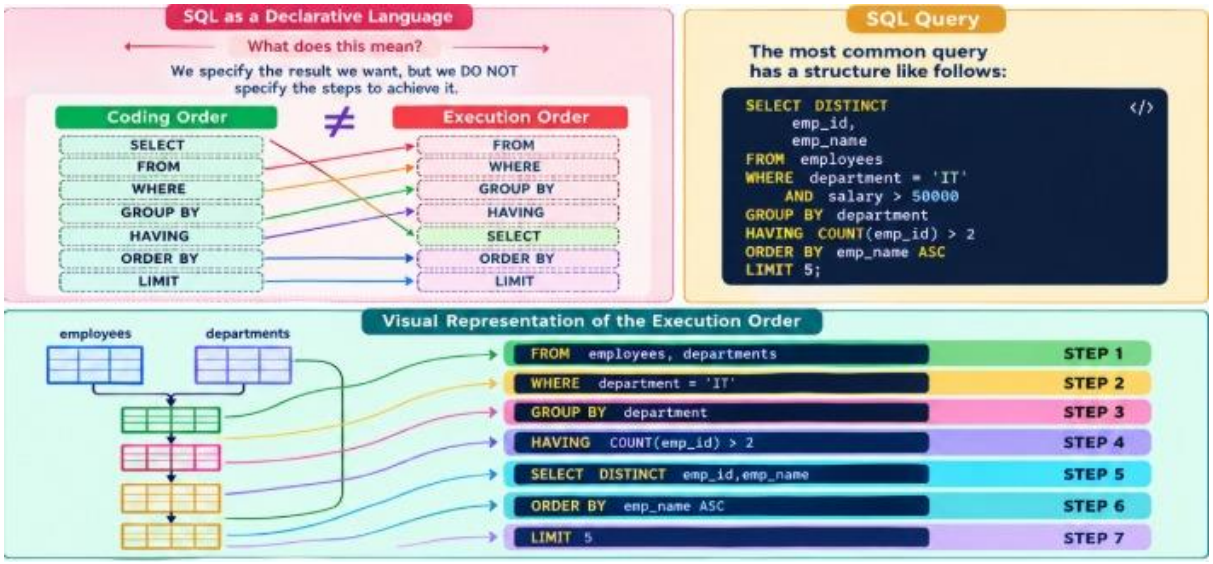
Objectives

- Perform SELECT, WHERE, ORDER BY, and DISTINCT queries.
- Search toys based on price, category, and availability.
- Retrieve customer and toy information.
- Apply filtering conditions.
- Generate basic business reports.

1. Basic SQL Queries

1.1 SELECT Query

The SELECT statement is used to retrieve data from a table.



Display all toys

```
SQL> SELECT * FROM PRODUCT;
```

```
PRODUCT_ID PRODUCT_NAME              CATEGORY_ID      PRICE      STOCK BRAND                                                                                                DESCRIPTION
---------- ------------------------- ----------- ---------- ---------- ---------------------------------------------------------------------------------------------------- ----------------------------------------
       101 ABC Learning Kit                    1        450         25 FunLearn                                                                                             Educational learning kit
       102 Teddy Bear                          2        550         20 ToyLand                                                                                              Soft teddy bear
       103 Remote Car                          3       1200         15 SpeedX                                                                                               Remote controlled car
       104 Chess Board                         4        750         18 Classic Games                                                                                        Wooden chess board
       105 Football                            5        450         30 PlayPro                                                                                              Outdoor football
       109 Superhero Figure                    9        600         16 HeroToys                                                                                             Action figure

6 rows selected.
```

Display selected toy details

```
SQL> SELECT Product_ID, Product_Name, Price FROM PRODUCT;
```

```
PRODUCT_ID PRODUCT_NAME                    PRICE
---------- ------------------------- ----------
       101 ABC Learning Kit                 450
       102 Teddy Bear                       550
       103 Remote Car                      1200
       104 Chess Board                      750
       105 Football                         450
       109 Superhero Figure                 600

6 rows selected.
```

1.2 WHERE Clause

The WHERE clause is used to filter records based on a condition.

Find toys with price less than 1000

```
SQL> SELECT * FROM PRODUCT
  2  WHERE Price < 1000;
```

```
PRODUCT_ID PRODUCT_NAME              CATEGORY_ID      PRICE      STOCK BRAND                                              DESCRIPTION
---------- ------------------------- ----------- ---------- ---------- -------------------------------------------------- --------------------------
------------------------------------------------------------------------------------------------------------------------------------------------------
       101 ABC Learning Kit                    1        450         25 FunLearn                                           Educational learning kit
       102 Teddy Bear                          2        550         20 ToyLand                                            Soft teddy bear
       104 Chess Board                         4        750         18 Classic Games                                      Wooden chess board
       105 Football                            5        450         30 PlayPro                                            Outdoor football
       109 Superhero Figure                    9        600         16 HeroToys                                           Action figure
```

Find toys with price greater than 100

```
SQL> SELECT * FROM PRODUCT
  2  WHERE Price > 100;
```

```
PRODUCT_ID PRODUCT_NAME              CATEGORY_ID      PRICE      STOCK BRAND                                              DESCRIPTION
---------- ------------------------- ----------- ---------- ---------- -------------------------------------------------- --------------------------
------------------------------------------------------------------------------------------------------------------------------------------------------
       101 ABC Learning Kit                    1        450         25 FunLearn                                           Educational learning kit
       102 Teddy Bear                          2        550         20 ToyLand                                            Soft teddy bear
       103 Remote Car                          3       1200         15 SpeedX                                             Remote controlled car
       104 Chess Board                         4        750         18 Classic Games                                      Wooden chess board
       105 Football                            5        450         30 PlayPro                                            Outdoor football
       109 Superhero Figure                    9        600         16 HeroToys                                           Action figure

6 rows selected.
```

1.3 ORDER BY

The ORDER BY clause is used to sort the query results.

Toys from lowest to highest price

```
SQL> SELECT * FROM PRODUCT
  2  ORDER BY Price ASC;
```

```
PRODUCT_ID PRODUCT_NAME              CATEGORY_ID      PRICE      STOCK BRAND                                              DESCRIPTION
---------- ------------------------- ----------- ---------- ---------- -------------------------------------------------- ----------------------------
       101 ABC Learning Kit                    1        450         25 FunLearn                                           Educational learning kit
       105 Football                            5        450         30 PlayPro                                            Outdoor football
       102 Teddy Bear                          2        550         20 ToyLand                                            Soft teddy bear
       109 Superhero Figure                    9        600         16 HeroToys                                           Action figure
       104 Chess Board                         4        750         18 Classic Games                                      Wooden chess board
       103 Remote Car                          3       1200         15 SpeedX                                             Remote controlled car

6 rows selected.
```

Toys from highest to lowest price

```
SQL> SELECT * FROM PRODUCT
  2  ORDER BY Price DESC;
```

```
PRODUCT_ID PRODUCT_NAME              CATEGORY_ID      PRICE      STOCK BRAND                                              DESCRIPTION
---------- ------------------------- ----------- ---------- ---------- -------------------------------------------------- ----------------------------
       103 Remote Car                          3       1200         15 SpeedX                                             Remote controlled car
       104 Chess Board                         4        750         18 Classic Games                                      Wooden chess board
       109 Superhero Figure                    9        600         16 HeroToys                                           Action figure
       102 Teddy Bear                          2        550         20 ToyLand                                            Soft teddy bear
       105 Football                            5        450         30 PlayPro                                            Outdoor football
       101 ABC Learning Kit                    1        450         25 FunLearn                                           Educational learning kit

6 rows selected.
```

1.4 DISTINCT

The DISTINCT keyword is used to display unique values.

Display unique categories

```
SQL> SELECT DISTINCT Category_ID
  2  FROM PRODUCT;
```

```
CATEGORY_ID
-----------
          1
          2
          3
          4
          5
          9

6 rows selected.
```

2. Search Products

2.1 Search Toys Based on Price

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE Price BETWEEN 400 AND 1000;
```

```
PRODUCT_ID PRODUCT_NAME              CATEGORY_ID      PRICE      STOCK BRAND                                              DESCRIPTION
---------- ------------------------- ----------- ---------- ---------- -------------------------------------------------- ----------------------------
       101 ABC Learning Kit                    1        450         25 FunLearn                                           Educational learning kit
       102 Teddy Bear                          2        550         20 ToyLand                                            Soft teddy bear
       104 Chess Board                         4        750         18 Classic Games                                      Wooden chess board
       105 Football                            5        450         30 PlayPro                                            Outdoor football
       109 Superhero Figure                    9        600         16 HeroToys                                           Action figure
```

2.2 Search Toys Based on Category

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE Category_ID = 1;
```

```
PRODUCT_ID PRODUCT_NAME            CATEGORY_ID      PRICE      STOCK BRAND                                                DESCRIPTION
---------- ----------------------- ----------- ---------- ---------- -------------------------------------------------- ----------------------------
------------------------------------------------------------------------------------------------------------------------------------------------------
       101 ABC Learning Kit                  1        450         25 FunLearn                                           Educational learning kit
```

This displays toys belonging to category ID 1

3. Retrieve Customer Information

3.1 Display All Customers

```
SQL> SELECT *
  2  FROM CUSTOMER;
```

```
        ID NAME                          AGE ADDRESS                          SALARY
---------- -------------------- ---------- ------------------------------ ----------
         1 Ramesh                       32 Ahmedabad                           30000
         2 Khilan                       25 Delhi                               30000
         3 Kaushik                      23 Kota                                30000
         4 Chaitali                     25 Mumbai                              75000
         5 Hardik                       27 Bhopal                              80000
```

Search for a Particular Customer

```
SQL> SELECT *
  2  FROM CUSTOMER
  3  WHERE ID = 1;
```

```
        ID NAME                          AGE ADDRESS                          SALARY
---------- -------------------- ---------- ------------------------------ ----------
         1 Ramesh                       32 Ahmedabad                           30000
```

4. Apply Filtering Conditions

Multiple conditions can be combined using AND, OR, and NOT.

4.1 AND Condition

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE Price < 1000
  4  AND Category_ID = 1;
```

```
PRODUCT_ID PRODUCT_NAME            CATEGORY_ID      PRICE      STOCK BRAND                                                DESCRIPTION
---------- ----------------------- ----------- ---------- ---------- -------------------------------------------------- ----------------------------
------------------------------------------------------------------------------------------------------------------------------------------------------
       109 Superhero Figure                  9        600         16 HeroToys                                           Action figure
```

4.2 OR Condition

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE Category_ID = 3
  4  OR Category_ID = 5;
```

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID	PRICE	STOCK	BRAND	DESCRIPTION
103	Remote Car	3	1200	15	SpeedX	Remote controlled car
105	Football	5	450	30	PlayPro	Outdoor football

4.3 IN Operator

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE Category_ID IN (1 , 2, 3);
```

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID	PRICE	STOCK	BRAND	DESCRIPTION
101	ABC Learning Kit	1	450	25	FunLearn	Educational learning kit
102	Teddy Bear	2	550	20	ToyLand	Soft teddy bear
103	Remote Car	3	1200	15	SpeedX	Remote controlled car

4.4 BETWEEN Operator

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE Price BETWEEN 100 AND 1000;
```

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID	PRICE	STOCK	BRAND	DESCRIPTION
101	ABC Learning Kit	1	450	25	FunLearn	Educational learning kit
102	Teddy Bear	2	550	20	ToyLand	Soft teddy bear
104	Chess Board	4	750	18	Classic Games	Wooden chess board
105	Football	5	450	30	PlayPro	Outdoor football
109	Superhero Figure	9	600	16	HeroToys	Action figure

5. Basic Business Reports

5.1 Product Price Report

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM PRODUCT
  3  ORDER BY Price DESC;
```

PRODUCT_ID	PRODUCT_NAME	PRICE
103	Remote Car	1200
104	Chess Board	750
109	Superhero Figure	600
102	Teddy Bear	550
105	Football	450
101	ABC Learning Kit	450

6 rows selected.

5.2 Customer Order Report

```
SQL> SELECT C.ID, C.NAME,
  2         O.ORDER_ID, O.ORDER_DATE,
  3         O.ORDER_STATUS, O.TOTAL_AMOUNT
  4  FROM CUSTOMER C
  5  JOIN ORDERS O
  6  ON C.ID = O.CUSTOMER_ID;
```

```
        ID NAME                 ORDER_ID ORDER_DAT ORDER_STATUS        TOTAL_AMOUNT
---------- -------------------- -------- --------- -------------------- ------------
         1 Ramesh                   1001 20-SEP-26 Confirmed                    1350
         2 Khilan                   1002 28-SEP-26 Shipped                       850
         3 Kaushik                  1003 22-SEP-26 Delivered                    1500
         4 Chaitali                 1004 23-SEP-26 Pending                       650
         5 Hardik                   1005 24-SEP-26 Confirmed                     950
         1 Ramesh                   1006 25-SEP-26 Shipped                      1100
         2 Khilan                   1007 26-SEP-26 Delivered                    1800
         3 Kaushik                  1008 27-SEP-26 Confirmed                     750

8 rows selected.
```

5.3 Product Order Report

```
SQL> SELECT P.Product_ID, P.Product_Name,
  2         OI.Order_ID, OI.Quantity, OI.Unit_Price
  3  FROM PRODUCT P
  4  JOIN ORDER_ITEM OI
  5  ON P.Product_ID = OI.Product_ID;
```

```
PRODUCT_ID PRODUCT_NAME               ORDER_ID   QUANTITY UNIT_PRICE
---------- ------------------------ ---------- ---------- ----------
       101 ABC Learning Kit               1001          2        300
       103 Remote Car                     1002          4        850
       104 Chess Board                    1003          1       1500
       105 Football                       1004          1        650
       109 Superhero Figure               1006          2        550
       101 ABC Learning Kit               1007          3        600
       103 Remote Car                     1008          1        750

7 rows selected.
```

5.4 Customer-wise Total Orders

```
SQL> SELECT C.ID, C.NAME,
  2         COUNT(O.ORDER_ID) AS TOTAL_ORDERS
  3  FROM CUSTOMER C
  4  LEFT JOIN ORDERS O
  5  ON C.ID = O.CUSTOMER_ID
  6  GROUP BY C.ID, C.NAME;
```

```
        ID NAME                 TOTAL_ORDERS
---------- -------------------- ------------
         1 Ramesh                          2
         2 Khilan                          2
         3 Kaushik                         2
         4 Chaitali                        1
         5 Hardik                          1
```

5.5 Customer-wise Total Purchase Amount

```
SQL> SELECT C.ID, C.NAME,
  2         SUM(O.TOTAL_AMOUNT) AS TOTAL_PURCHASE
  3  FROM CUSTOMER C
  4  JOIN ORDERS O
  5  ON C.ID = O.CUSTOMER_ID
  6  GROUP BY C.ID, C.NAME;
```

```
        ID NAME                 TOTAL_PURCHASE
---------- -------------------- --------------
         1 Ramesh                         2450
         2 Khilan                         2650
         3 Kaushik                        2250
         4 Chaitali                        650
         5 Hardik                          950
```